SETH FRIEND

726-234-4817 | seth.friend@utexas.edu | sethfriendportfolio | humanoid demo | linkedin.com/in/seth-friend

Education

University of Texas at Austin *Austin, TX*
Integrated B.S./M.S. in Electrical and Computer Engineering | GPA: 3.79 *B.S. May 2027; M.S. May 2028*
- Relevant Coursework: Human Centered Robotics, Algorithms, Embedded Systems, Honors Probability, Controls, Digital Signal Processing

Robotics Projects

Humanoid Robot | *Onshape, Python, Isaac Lab, MuJoCo, CAN* *July 2025 – Present*
- Designed and built an 8 kg, 20-DoF humanoid with a sub-$2K component cost, integrating mechanical design, electronics, actuators, and URDF/USD simulation models
- Achieved untethered hardware walking over approximately 10 feet using PPO policies trained in Isaac Lab across 4,096 parallel environments
- Built a 50 Hz Raspberry Pi control stack with CAN actuator control, IMU feedback, calibration, and safety checks
- Validated policies in MuJoCo via sim-to-sim transfer, evaluating robustness to physics-engine differences
- Improved gait posture, reducing forward lean from 14.5° to 3–5° and toe-out from up to 23° to 3–4°

Experience

Technical Assistant - Former Design Lead *Austin, TX*
Texas Inventionworks | Design and Digital Infrastructure Teams *Aug 2024 – Present*
- Guide multidisciplinary student projects through embedded-system development, fabrication, prototyping, and design workflows in a university makerspace
- Support hands-on troubleshooting and iteration across mechanical, electrical, and software project domains
- Built a usage-analytics dashboard for Texas Inventionworks' staff portal in JavaScript, turning Fabman access-control logs for 14 machines into an hour-by-weekday heatmap of peak demand

Systems Engineering Intern - Hardware-in-the-Loop Simulation *Phoenix, AZ*
Honeywell Aerospace | Validation Facility *June – Aug 2026*
- Integrated 16 new signals into a Cessna Citation Sovereign simulation bench containing 1,000+ wires, completing physical wiring, software configuration, and end-to-end testing
- Debugged and corrected electrical and software-configuration faults across hardware-in-the-loop benches, helping restore system functionality and test readiness
- Executed acceptance test procedures on a Bombardier bench, validating 500+ signals connected to real avionics hardware
- Built a Python pipeline that parsed SSI configuration files and generated user-ready Excel workbooks, eliminating manual formatting for test documentation

Lead Electromechanical Technician - Amazon Robotics Deployment *Austin, TX*
Ivy Technology *May – July 2025*
- Led teams of 7–10 through low-volume assembly of 12 Cardinal automated work cells, completing the deployment one week ahead of schedule
- Installed and wired 400+ automation-ready chutes while coordinating mechanical and electrical integration with other leads
- Performed quality assurance and hands-on troubleshooting on 20 Linux-based autonomous mobile robots

Machine Learning Researcher and Robotics Mentor *Austin, TX*
Robot Interactive Intelligence Lab *Jan – Dec 2024*
- Integrated motion-capture input with a robot-control workflow and configured Ubuntu/ROS environments
- Trained a simulated robot arm with robomimic to achieve a 95% task success rate hitting a balloon and mentored 20+ students in ROS, PyTorch, and kinematics

Technical Skills

Programming: Python, C++, C, Bash
Robotics and Simulation: Isaac Lab, MuJoCo, PyTorch, reinforcement learning, URDF/USD, ROS
Hardware and Design: Onshape, electromechanical integration
Tools and Analysis: Linux, Git, NumPy, pandas, Matplotlib